

Display Simulation Model-based Backlight Optimization and Compensation for Enhanced Picture Quality FALD Displays

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Abstract

Full-array local dimming displays have been considered highly desirable and promising across diverse applications, such as automotive digital clusters, cockpit displays, home appliances, and advertisements. However, despite the high potential to deliver a high dynamic range of picture quality, previous local dimming optimization and compensation algorithms limit the picture quality of full-array local dimming displays mainly due to halo artifacts. This paper proposes the display simulation model-based compensation algorithm as well as the backlight optimization algorithms for full-array local dimming displays. Experimental results on three full-array local dimming displays show that the proposed method outperforms the compared methods in qualitative/quantitative picture quality and power saving.

Author Keywords

full-array local dimming; display simulation model; halo artifacts; high contrast; high dynamic range display; backlight optimization

1. Introduction

Full array local dimming displays (FALD) have been considered highly desirable and promising in diverse applications: automotive cockpit displays, digital clusters, infotainment displays, advertisement, and home appliances [1]. FALDs are capable of delivering high contrast by selectively dimming backlight zones in darker areas, resulting in significantly enhanced, deeper blacks and improved overall picture quality. Despite high contrast and superior picture quality, FALDs suffer from undesirable artifacts such as halos and luminance non-uniformity. Several studies have been conducted to improve the viewing experience and deal with the artifacts, including backlight optimization algorithms [2, 3, 4, 5, 6, 7, 8] and compensation algorithms [9, 10].

Picture quality of FALDs is affected by the combination of the backlight optimization algorithm, the displayed image, and the backlight characteristics [11, 12]. Despite its promising potential to deliver high dynamic range picture quality, there is a lack of compensation methods for FALDs to address halo artifacts and inconsistent luminance values across different content and backlight zones [11]. Thus, it is crucial to develop effective compensation methods for practical FALD systems to be successful.

This paper proposes an effective and efficient framework to obtain high picture quality while reducing the power consumption of FALD systems. The main contributions of this paper are summarized as follows.

- This paper proposes a content-adaptive and luminance-consistent backlight optimization method, which provides high contrast and brightness consistency across different images.
- This paper proposes a display simulation model-based

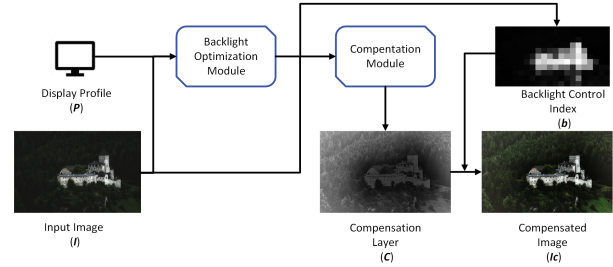


Figure 1: The block diagram of the proposed method

compensation algorithm, which effectively compensate halo artifacts and increases luminance uniformity.

- Qualitative and quantitative experimental results prove that the proposed method achieves high picture quality while significantly reducing power consumption in FALD systems.

The remainder of this paper is organized as follows. Section 2 describes the proposed display simulation model-based backlight optimization and compensation algorithm in detail. Experimental results are analyzed in Section 3. Lastly, Section 4 concludes this paper with some future works.

2. Proposed Method: Display Model-based Backlight Optimization and Compensation

The backlight optimization and compensation algorithms play a crucial role in delivering high-quality pictures in an FALD system. This paper presents a display simulation model-based backlight optimization and compensation algorithm as an advanced variant of [1, 9, 11].

As illustrated in Fig. 1, the proposed method mainly consists of two modules: the backlight optimization module and the compensation module. The proposed method takes an image, I , and a display profile, P , as input. The display profile, P , includes the dimensions of the backlight and the point spread function (PSF) profile of one backlight LED.

Backlight Optimization Module: The backlight optimization module generates a backlight control index, b , based on the luminance of the input image I . The input image, I , is first divided into grids, with each grid, $g_{i \in N_z}$, centered at the location of a backlight LED, where i indicates the i^{th} backlight zone and N_z indicates the number of zones. The maximum luminance value is selected as the backlight control index for each grid, b_i . The generated backlight control index is normalized into (0 - 1) to obtain b_n . Then, b_n is multiplied by the brightness consistency control value, s_c , to generate the backlight control index, b , which delivers consis-



Figure 2: Examples of intermediate results. (a): input image, $\mathbf{I}$, (b): diffused backlight map, $\mathbf{B}$, (c): perceptual image for the FALD display, $\mathbf{I}_L$, (d): perceptual image for the ideal display, $\mathbf{I}_I$, (e): perceptual image for the compensated image on the FALD display, $\mathbf{I}_P$, (f): compensated image, $\mathbf{I}_c$

tent brightness across different images. The brightness consistency control value, s_c is defined as $s_c = b_t / \max \mathbf{b}_n$, where b_t represents the target brightness of the display.

Compensation Module: Display Models and Compensation Algorithm: The compensation module takes an image ($\mathbf{I}$), display profile ($\mathbf{P}$), and the generated backlight control index ($\mathbf{b}$) as input and generates the compensation layer ($\mathbf{C}$). The compensation layer, $\mathbf{C}$, is combined with the input image, $\mathbf{I}$, to compensate for potential halo artifacts and inconsistent luminance/color in different zones. In the compensation module, the diffused backlight map, $\mathbf{B}$, is first generated as follows:

$$\mathbf{B}_p = \sum_{i \in N_z} \mathbf{b}_i \cdot \mathcal{P}(\|\mathbf{p} - \mathbf{z}_i\|_2) \quad (1)$$

where $\mathbf{B}_p$ is the diffused backlight map value at the pixel, $\mathbf{p}$, and $\mathcal{P}$ represents the PSF of each LED. $\mathbf{b}_i$ and $\mathbf{z}_i$ represent the i^{th} backlight control index and the center location of the i^{th} backlight LED zone, respectively. $\|\cdot\|_2$ indicates the L2 norm. The diffused backlight map, $\mathbf{B}$, is then used to generate the perceptual image for the FALD display, $\mathbf{I}_L$, based on the FALD display model as follows:

$$\mathbf{I}_{L,k} = \mathbf{I}_k^\gamma \cdot \mathbf{B} + L_{bl} + \kappa \cdot ((1 - \mathbf{I}_k^\gamma) \cdot \mathbf{B} + L_{bl}) \quad (2)$$

where the subscript $k \in (R, G, B)$ represents the color channel. γ is the sRGB transfer parameter, determined by the display characteristic (2.2 as the default). L_{bl} and κ indicate the black level luminance of the FALD display in cd/m^2 and the display light leakage, respectively. Then, the perceptual image for the ideal display, $\mathbf{I}_I$, is generated as follows:

$$\mathbf{I}_{I,k} = \mathbf{I}_k^\gamma \cdot \mathbf{1} + L_{bi} \quad (3)$$

where $\mathbf{1}$ indicates an all-one matrix, and L_{bi} indicates the black level luminance of the ideal display in cd/m^2 . Note that the ideal display model, (3), can be derived from the FALD display model, (2), by setting the display light leakage, κ , to zero, and the resolution of the backlight, $\mathbf{b}$ to be the same as the display resolution. Based on the FALD perceptual image and the ideal perceptual image, the compensation layer, $\mathbf{C}$ is computed as follows:

$$\mathbf{C} = \frac{\mathbf{I}_I}{\mathbf{I}_L + \varepsilon} \quad (4)$$

where ε is an adopted small constant value to mitigate potential divide-by-zero errors. Then, the computation layer, $\mathbf{C}$, is combined with the input image, $\mathbf{I}$, to generate the compensated image $\mathbf{I}_c$, as follows:

$$\mathbf{I}_c = \mathbf{I} \cdot \mathbf{C} = \mathbf{I} \cdot \frac{\mathbf{I}_I}{\mathbf{I}_L + \varepsilon} \quad (5)$$

The compensated image, $\mathbf{I}_c$, is displayed on FALD displays where the generated backlight control index, $\mathbf{b}$, controls backlight LEDs. Examples of intermediate results from the proposed algorithm are shown in Fig. 2.

3. Experimental Results

Experimental Setups : In order to evaluate the performance of the proposed method, it is compared with five different algorithms: backlight optimization and compensation, on three FALD displays: FALD#1 with 252 zones and 2560x1440 image resolution, FALD#2 with 200 zones and 3840x2160 image resolution, and FALD#3 with 270 zones and 1920x720 image resolution.

In the experiments, 26 original images were adopted and processed with different local dimming algorithms (LDA). The proposed method is compared with 5 different LADs. LDA1 is a high-contrast backlight optimization with a pixel-wise compensation, and LDA2 is a low-contrast backlight optimization with a pixel-wise compensation. LDA3 is a medium-contrast backlight optimization without a compensation algorithm, and LDA4 is a high-contrast backlight optimization without a compensation algorithm. LDA5 is full-backlight-on backlight optimization without a compensation algorithm. Due to high-contrast backlight optimization, LDA1 and LDA4 suffer severe halo artifacts. The quantitative picture quality is evaluated in terms of the perceptual contrast, halo artifacts, luminance uniformity, and brightness consistency. In addition, power consumption is also analyzed. Radiant ProMetric I29 Imaging Photometer [13] and XRITE i1Pro [14] were used to capture the luminance of the displayed image on FALD displays.

Quantitative Evaluation - Metrics : As recommended in [11], the global contrast ratio, C_g , and local contrast ratio, C_l , are used to measure the perceptual contrast of FALD displays, which are defined as follows:

$$C_g = \frac{L_{max} + \varepsilon}{L_{min} + \varepsilon} \quad (6)$$

$$C_l = \frac{1}{N_r} \sum_j \frac{L_b + \varepsilon}{L_d + \varepsilon} \quad (7)$$

where L_{max} and L_{min} represent the maximum and minimum luminance values within the entire image, respectively. N_r represents the number of selected regions. L_b and L_d represent the brightest and darkest luminance values within each region, respectively.

For halo artifact evaluation, the halo size and halo luminance are used, which are measured on a specifically designed pattern image,

Table 1: Evaluation of the proposed method compared with different LDAs on FALD#1, FALD#2, and FALD#3. $\mathcal{P}$ represents power consumption. Higher values indicate better for C_g and C_l while lower values indicate better for ξ , Λ , U_z , U_c , and $\mathcal{P}$.

	C_g	C_l	ξ	Λ	U_z	U_c	$\mathcal{P}$ (mW/nit)
FALD#1							
LDA1	1201.6	41.3	1.72	0.20%	8.6%	-	17
LDA2	670.9	21.2	1.70	0.10%	10.3%	-	21
ours	1501.7	56.1	0.85	0.06%	4.4%	-	12
FALD#2							
LDA3	367.9	24.5	-	-	-	15.6%	93
LDA4	481.1	25.1	-	-	-	20.1%	62
Ours	534.5	30.5	-	-	-	6.5%	58
FALD#3							
LDA5	841.1	58.5	-	-	-	-	27
Ours	2237.2	80.6	-	-	-	-	12

where the white square of the backlight zone size spans two different zones [11]. The halo size, ξ , is defined as the relative size of the halo to the backlight zone size as follows:

$$\xi = \frac{dH}{X} \quad (8)$$

where X and dH represent the size of the backlight zone and the size of the halo, respectively. The halo luminance, Λ , is defined as the relative luminance of the halo to the peak luminance at the center of the zone, which is defined as follows:

$$\Lambda = \frac{L_{zb}}{L_p} \quad (9)$$

where L_{zb} and L_p represent the luminance of the halo at the zone boundary and the peak luminance within the backlight zone, respectively. Please refer to [11] for more details.

In order to measure the luminance uniformity across different zones within an image, U_z is employed, defined as:

$$U_z = 100\% \frac{\sqrt{\sum_x^{N_x} (L_x - L_m)^2}}{L_m} \quad (10)$$

$$L_m = \frac{1}{N_x} \sum_x^{N_x} L_x \quad (11)$$

where L_x and L_m represent the luminance value for each measured point, x , and the average luminance value of all the measured points. N_x is the number of the measured points.

The content across luminance uniformity, U_c , is used to evaluate the brightness consistency, which measures how the maximum brightness varies across different contents.

$$U_c = 100\% \frac{\sqrt{\sum_y^{N_y} (\hat{L}_y - \hat{L}_m)^2}}{\hat{L}_m} \quad (12)$$

$$\hat{L}_m = \frac{1}{N_y} \sum_y^{N_y} \hat{L}_y \quad (13)$$

where $\hat{L}_y$ and $\hat{L}_m$ represent the max-luminance value for each image, y , and the average max-luminance value of all the images. N_y is the number of images.

Quantitative Evaluation - Results : Table 1 shows the analysis comparison of the proposed method with other algorithms on three FALDs: FALD#1, FALD#2, FALD#3. As shown in Table 1, the proposed method shows higher contrast on all three different FALDs. On average, 72% of the local contrast and 126% of the global contrast were increased. Also, the proposed method significantly reduced the halo size and the halo luminance compared to LDA-1 and LDA-2 on FALD#1. On average, 70% of halo size, ξ , and 50% of halo luminance, Λ , were reduced. Further, the proposed method decreases U_z by 5% and U_c by 11.3% on FALD#1 and FALD#2, respectively, demonstrating that it provides higher brightness and greater luminance consistency.

In addition to the picture quality measurements, we also analyzed the power consumption of different algorithms. As shown in Table 1, the proposed method consumed the least power, achieving an average of 35% power savings compared to the compared algorithms. In summary, the proposed method provides superior picture quality compared to other algorithms in terms of contrast, halos, luminance uniformity, and brightness consistency, while reducing power consumption.

Qualitative Evaluation : In addition to the quantitative evaluation, this paper qualitatively compares the picture quality of the proposed method. Fig. 3 depicts the luminance image of the proposed method and the compared algorithm captured by the Radiant imaging sensor [13] on FALD#1 display, showing that the perceptual quality of the proposed method is superior to the compared algorithm.

4. Conclusion

This paper proposes display simulation-based backlight optimization and compensation algorithms. Quantitative and qualitative analysis on diverse FALD displays showed outperforming results of the proposed method in contrast ratio, halo artifact reduction, brightness consistency, luminance consistency, and power consumption. Future works will include color space compensation and an on-the-fly display profiling algorithm for enhanced practical usage.

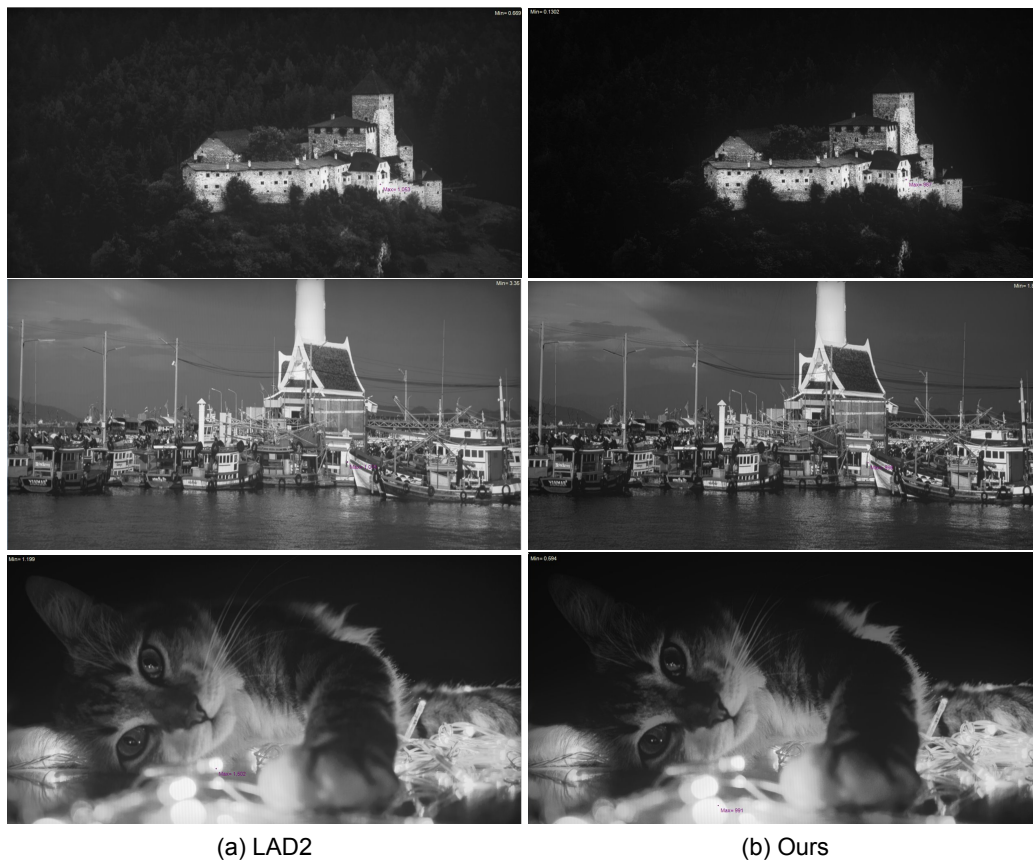


Figure 3: Qualitative evaluation of the proposed method. Luminance images are captured by the imaging sensor, which was displayed on FALD#1. (a) The left columns are LAD2, and (b) the right columns are the proposed method.

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